

STATEMENT OF WORK

TO:
MDF SERVER ROOM COOLING SYSTEM REPLACEMENT

AT:
KAYENTA HEALTH CARE CENTER
KAYENTA, ARIZONA

DATE:
NOVEMBER 19, 2025

PERIOD OF PERFORMANCE:
AWARD DATE – DECEMBER 31, 2026

INTRODUCTION

Kayenta Health Center (KHC) operates a critical Main Distribution Frame (MDF) server room that supports network, telecommunications, and medical data systems essential to healthcare operations. The existing Liebert precision cooling system has reached the end of its service life and no longer meets capacity requirements due to planned IT and server expansions. This project provides for a complete replacement and upgrade of the precision cooling systems to ensure reliable, year-round temperature and humidity control in compliance with federal, ASHRAE, and KHC facility standards.

BACKGROUND

The MDF room relies on dual Liebert PFH037A-AH7 units operating simultaneously to maintain a setpoint temperature of 20°C (68°F). During cold weather, the rooftop condensing units experience low-head pressure issues that reduce cooling performance. These issues, combined with increased server density, necessitate replacement with modern precision cooling systems featuring low-ambient operation and N+1 redundancy.

OBJECTIVE

Replace the existing cooling systems with two (2) new six-ton Vertiv Liebert precision cooling systems, configured for N+1 redundancy and equipped with integrated humidity control, energy-efficient operation, and low-ambient capability rated to -20°F (-29°C).

PERFORMANCE DESCRIPTION

The Contractor shall furnish all labor, materials, tools, equipment, supervision, and transportation to remove the existing systems and install new precision cooling equipment. All work shall be performed in accordance with ASHRAE TC 9.9, ASHRAE 90.1, DOE 10 CFR 431, GSA P-100, EPA Section 608, and NEC (NFPA 70). Work shall be coordinated with Facility Maintenance to minimize disruption.

SCOPE OF WORK

1. Recover refrigerant and remove existing indoor and outdoor units per EPA Section 608.
2. Furnish and install two (2) Vertiv Liebert six-ton precision cooling systems in N+1 configuration.
3. Provide outdoor condensing units rated for low-ambient operation (-20°F minimum) with variable-speed condenser fans.
4. Install indoor ceiling-mounted air handlers with integrated humidifiers and condensate drains.
5. Replace refrigerant piping, insulation, and supports; pressure test and evacuate per manufacturer requirements.
6. Verify and replace electrical disconnects, wiring, and breakers to meet NEC.
7. Integrate Liebert iCOM or approved equal controls for environmental monitoring and redundancy coordination.
8. Conduct full startup, testing, and commissioning with factory-certified technicians.
9. Provide as-built drawings, O&M manuals, and training upon project completion.

COLD WEATHER PERFORMANCE REQUIREMENTS

The system shall include features for reliable operation during low-ambient conditions:

- Low-ambient control package with stable operation to -20°F (-29°C).
- Variable-speed fans or fan cycling for head pressure control.
- Flooded condenser or equivalent head pressure management.
- Crankcase heaters and coil guards for rooftop applications.
- Field verification of head pressure stabilization during cold-weather commissioning.

HUMIDITY CONTROL AND DEHUMIDIFICATION

The precision cooling systems shall include integrated humidity control capable of both humidification and dehumidification to maintain 40–60% RH. Dehumidification shall occur via the cooling coil and reheat process, while humidification shall be achieved through electrode steam or infrared systems. Control shall be automatic via the Liebert iCOM to maintain 20°C ± 2°C and 45% ± 5% RH year-round.

COMPLIANCE REQUIREMENTS

- ASHRAE TC 9.9 – Thermal Guidelines for Data Processing Environments
- ASHRAE 90.1 and 62.1 – Energy and Ventilation Standards
- DOE 10 CFR 431 – Federal Energy Efficiency Standards
- GSA P-100 – Facilities Standards for Public Buildings
- Joint Commission EC.02.01.01, EC.02.05.01, and EC.02.05.07
- NEC (NFPA 70), EPA Section 608, and OSHA 29 CFR 1910

CONTRACTOR QUALIFICATIONS

All work shall be performed by licensed HVAC and electrical contractors with a minimum of three (3) comparable projects within the last five (5) years. All technicians shall hold valid EPA Section 608 certification for refrigerant handling.

REPORTING AND DOCUMENTATION

Contractor shall provide submittals for review and approval, daily construction logs, a safety plan, as-built drawings, and final testing reports.

TESTING AND COMMISSIONING

Final commissioning shall confirm temperature, humidity, and redundancy performance under supervision of the Facility Representative. The contractor shall submit signed verification reports prior to acceptance.

QUALITY CONTROL AND SAFETY

The Contractor shall follow KHC safety and infection control protocols, maintain a site-specific quality plan, and ensure all personnel carry visible identification and PPE. Work shall be coordinated to minimize patient and staff disruption.

ENVIRONMENTAL AND SITE RESTORATION

All disturbed areas, including roof penetrations, shall be restored to equal or better condition. Any roofing cuts or penetrations for piping or conduit shall be properly sealed and flashed per manufacturer specifications. All debris shall be removed daily.

GENERAL REQUIREMENTS

Work hours: 8:00 AM – 5:00 PM, Monday through Friday, excluding Federal Holidays. Alternate hours require Facility Representative approval. All areas shall be kept clean and restored after work completion.

PROJECT CLOSEOUT

Provide training for maintenance staff, warranty documentation, performance data, and commissioning results upon completion.

CONTACT INFORMATION

Desmond Jones – Supervisory General Engineer – (928) 697-5192

Vernon Begay – Maintenance Mechanic Supervisor – (928) 697-5194

Kendall Tanner – Maintenance Mechanic Supervisor – (928) 697-5193

WARRANTY

Workmanship Warranty: One (1) year or better.

Equipment Warranty: One (1) year minimum, five (5) years compressor warranty. Contractor shall guarantee full operational functionality and compliance upon final acceptance.



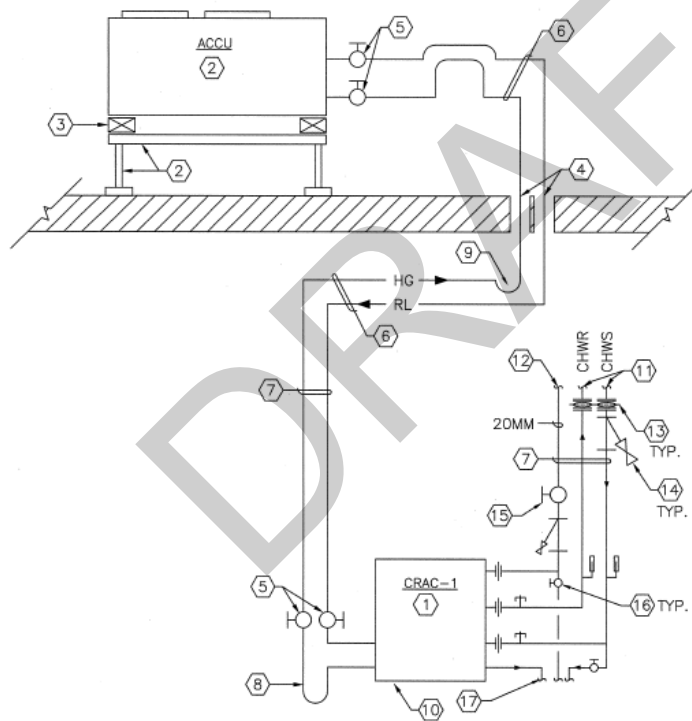












KEYED NOTES:

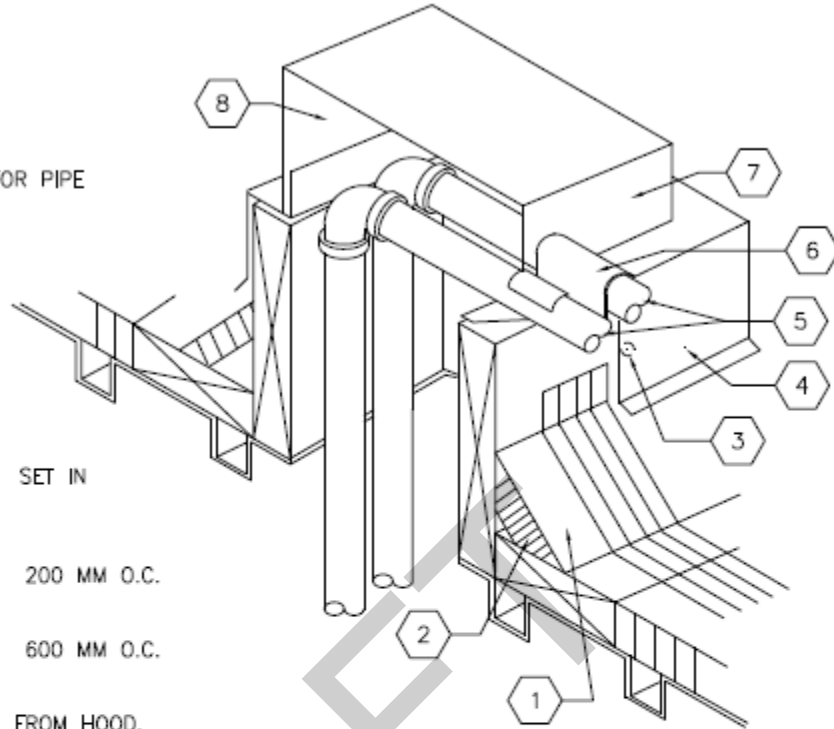
- 1 COMPUTER ROOM A/C UNIT, DUAL COOLING SYSTEM WITH CHILLED WATER FROM CENTRAL PLANT (PRIMARY) AND DX (BACKUP). SEE EQUIPMENT SCHEDULE.
- 2 ROOF MOUNTED AIR COOLED CONDENSER WITH 18" THYBAR TYPE SUPPORT BASE. SEE MECHANICAL EQUIPMENT SCHEDULE AND ARCHITECTURAL/STRUCTURAL FOR SUPPORT BASE AND FLASHING.
- 3 12MM FIBERGLASS VIBRATION ISOLATION PADS, TYP.
- 4 ROOF PIPING PENETRATIONS, SEE DETAIL
C4
M504 M503
- 5 REFRIGERANT MANUAL SHUTOFF VALVES PER MANUFACTURER RECOMMENDATIONS.
- 6 PITCH HORIZONTAL REFRIGERANT PIPING IN DIRECTION OF REFRIGERANT FLOW A MINIMUM OF 12MM IN 3M FEET.
- 7 ALL PIPING VALVES AND FITTINGS EXPOSED WITHIN THE ROOM BEING SERVED, INCLUDING CHILLED WATER, COLD WATER MAKEUP, REFRIGERATION LIQUID AND HOT GAS DISCHARGE, SHALL BE INSULATED AND COVERED WITH ALUMINUM JACKET. SEE SECTION 15250 FOR ADDITIONAL INSULATION REQUIREMENTS.
- 8 PROVIDE TRAP FOR HOT GAS DISCHARGE PIPING PER MANUFACTURER'S RECOMMENDATIONS.
- 9 PROVIDE INVERTED TRAPS FOR HG AND RL PIPING IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS. TRAP TO EXTEND ABOVE TOP OF CONDENSER COIL A MINIMUM OF 190MM.
- 10 PROVIDE UNIT WITH OVERFLOW SWITCH TO SHUT DOWN UNIT PRIOR TO DRAIN OVERFLOW AND SEND ALARM SIGNAL TO EMCS.
- 11 CHS AND CHR FROM CHILLED WATER MAINS. SEE FLOOR PLANS FOR PIPE SIZE.
- 12 12MM NON-POTABLE COLD WATER MAKEUP TO UNIT HUMIDIFIER, FROM RPB PREVENTER. SEE PLUMBING DRAWINGS FOR CONTINUATION.
- 13 BALL VALVE.
- 14 STRAINER.
- 15 BALL VALVE.
- 16 20MM DRAIN VALVE WITH HOSE CONNECTION AND CAP.
- 17 20MM CONDENSATE DRAIN FROM UNIT TO FLOOR SINK WITH INDIRECT CONNECTION.

COMPUTER ROOM A/C UNIT SCHEMATIC
D5
SCALE = NONE
M103G
M701
M111G
M112

NOTE: SEE PLANS FOR PIPE
SIZE.

KEYED NOTES:

- 1 BASE FLASHING
- 2 FIBER CANT STRIP— SET IN
BITUMEN.
- 3 FASTENERS APPROX. 200 MM O.C.
- 4 FASTENERS APPROX. 600 MM O.C.
- 5 SLOPE PIPES AWAY FROM HOOD.
- 6 SHEET METAL OR FLEX— TUBE
COLLAR.
- 7 SHEET METAL HOOD.
- 8 INSULATE INSIDE OF METALWORK IN
COLD CLIMATES.



PIPING THRU ROOF DETAIL

SCALE = NONE